Taipei Veterans General Hospital

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Sample Information

Patient Name: 洪苡溱
Gender: Female
ID No.: A224879886
History No.: 46338005
Age: 43

Ordering Doctor: DOC5636D 吳紋綺
Ordering REQ.: 0BWTMGW
Signing in Date: 2022/06/23

Path No.: S111-99612
MP No.: F22064
Assay: Oncomine Focus Assay
Sample Type: FFPE
Block No.: S111-76392B
Percentage of tumor cells: 90%

Reporting Doctor: DOC5466K 葉奕成 (Phone: 8#5466)

Note:

Sample Cancer Type: Melanoma

Table of Contents

	Page
Variant Details	2
Biomarker Descriptions	2
Relevant Therapy Summary	3
Relevant Therapy Details	4
Clinical Trials Summary	5
Alert Details	5

Report Highlights

2 Relevant Biomarkers
1 Therapies Available
1 Clinical Trials

Relevant Melanoma Variants

Gene	Finding
BRAF	None detected
KIT	***KIT p.(L576P) c.1727T>C, KIT amplification***
NRAS	None detected
NTRK1	None detected
NTRK2	None detected
NTRK3	None detected

Disclaimer: The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2022.05(005).The content of this report has not been evaluated or approved by FDA, EMA or other regulatory agencies.

Relevant Biomarkers

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
IA	*KIT p.(L576P) c.1727T>C* KIT proto-oncogene, receptor tyrosine kinase Allele Frequency: 43.38%	imatinib	imatinib	0
IIC	*BAG4-FGFR1 fusion* BAG cochaperone 4 - fibroblast growth factor receptor 1	None	None	1

Public data sources included in relevant therapies: FDA[1], NCCN, **EMA**[2], ESMO
Tier Reference: Li et al. *Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists.* J Mol Diagn. 2017 Jan;19(1):4-23.

Prevalent cancer biomarkers without relevant evidence based on included data sources

KIT amplification

Variant Details

DNA Sequence Variants

Gene	Amino Acid Change	Coding	Variant ID	Locus	Allele Frequency	Transcript	Variant Effect	Coverage
KIT	p.(L576P)	c.1727T>C	COSM1290	chr4:55593661	43.38%	NM_000222.3	missense	1948
FGFR4	p.(P136L)	c.407C>T	.	chr5:176517797	99.70%	NM_213647.3	missense	1999
RET	p.(L769=)	c.2307G>T	.	chr10:43613843	100.00%	NM_020975.6	synonymous	1999

Gene Fusions (RNA)

Genes	Variant ID	Locus	Read Count
BAG4-FGFR1	BAG4-FGFR1.B1F2	chr8:38034657 - chr8:38315052	753

Copy Number Variations

Gene	Locus	Copy Number
KIT	chr4:55529117	7.41

Biomarker Descriptions

FGFR1 (fibroblast growth factor receptor 1)

Background: The FGFR1 gene encodes fibroblast growth receptor 1, a member of the fibroblast growth factor receptor (FGFR) family that also includes FGFR2, 3, and 4. These proteins are single transmembrane receptors composed of three extracellular immunoglobulin (Ig)-type domains and an intracellular kinase domain. Upon FGF-mediated stimulation, FGFRs activate several oncogenic signaling pathways, including the RAS/RAF/MEK/ERK, PI3K/AKT/MTOR, PLC/PKC, and JAK/STAT pathways influencing cell proliferation, migration, and survival[1,2,3].

Alterations and prevalence: Recurrent somatic alterations common to the FGFR family include gene amplification, mutation, and chromosomal translocations leading to FGFR fusions[4]. Amplification of FGFR1 is observed in 15-20% of squamous lung cancer, 10-15% of breast cancer, 8% of bladder cancer, and 2-5% of uterine cancer cases[5,6,7,8,9]. The most common recurrent mutations, N546K and K656E, are relatively infrequent (<1%); they activate mutations in the kinase domain and are distributed in diverse cancer

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Biomarker Descriptions (continued)

types[10]. FGFR1 translocations giving rise to expressed fusions are common in certain hematological cancers, but less common in solid tumors[11,12,13].

Potential relevance: The FDA has granted fast-track designation (2018) to Debio 1347[14] for solid tumors harboring aberrations in FGFR1, FGFR2, or FGFR3. FDA has approved multi-kinase inhibitors, including regorafenib, ponatinib, lenvatinib, nintedanib, and pazopanib, that are known to inhibit FGFR family members. These inhibitors have demonstrated anti-tumor activity in select cancer types with FGFR alterations[15,16,17,18,19,20,21]. In a phase II clinical trial, dovitinib, a multi-tyrosine kinase inhibitor (TKI), exhibited an overall response rate (ORR) of 11.5% and a disease control rate (DCR) of 50% in patients with advanced squamous cell lung cancer possessing FGFR1 amplification. The patients had a median overall survival (OS) of 5 months and progression-free survival (PFS) of 2.9 months[22]. Likewise, in a phase Ib study testing the FGFR inhibitor AZD4547, median OS was 4.9 months in patients with FGFR1-amplified advanced squamous cell lung cancer. One of 13 (8%) patients achieved a partial response, 4 (31%) exhibited stable disease, and 2 (13.3%) demonstrated PFS at 12 weeks[23].

KIT (KIT proto-oncogene, receptor tyrosine kinase)

Background: The KIT gene, also known as CD117, encodes the KIT proto-oncogene receptor tyrosine kinase (c-KIT), a member of the PDGF receptor type III receptor tyrosine kinase family, which includes PDGFRA, PDGFRB, CSF1R, FLT1, FLT3, FLT4 and KDR[24,25]. KIT is a receptor for stem cell factor, important in regulating growth and development of hematopoietic cells[26]. The KIT gene is flanked by the PDGFRA and KDR genes on chromosome 4q12. Ligand binding to KIT results in kinase activation and stimulation of downstream pathways including the RAS/RAF/MEK/ERK and PI3K/AKT/MTOR pathways promoting cell proliferation and survival[27].

Alterations and prevalence: Recurrent somatic KIT alterations are observed in both solid and hematological cancers and include activating mutations such as single nucleotide variants, small duplications, and complex in-frame insertions or deletions (indels). Mutations in KIT exons 8, 9, 11, and 17 disrupt auto-inhibitory mechanisms and lead to constitutive activity[28]. Gain of function mutations are found in up to 70% of mast cell tumors, 17% of nasal T-cell lymphomas, and 9% of dysgerminoma[29]. Somatic mutations in exon 11 occur in 60-70% of all gastrointestinal stromal tumor (GIST), whereas alterations in exons 8 and 17 are more common in myeloid cancers[9,28,29]. A common kinase domain mutation that causes ligand-independent constitutive activation, D816V, occurs in 80-93% of aggressive forms of mastocytosis[30,31].

Potential relevance: Imatinib[32] (2001) is approved for KIT positive malignant GIST and adult patients with aggressive systemic mastocytosis (SM) harboring D816V mutations. Imatinib is also recommended for KIT activating mutations in melanoma and exon 9 and 11 mutations in GIST[33,34,35]. Mutations in exon 17 have been identified to confer resistance to imatinib and sunitinib[36]. Patients with acute myeloid leukemia (AML) that harbor KIT activating mutations with t(8;21) and inv(16) have an increased risk of relapse[37]. KIT D816V mutation is associated with the diagnosis of SM and aggressiveness of the disease[38,39].

Relevant Therapy Summary

● In this cancer type ○ In other cancer type ◑ In this cancer type and other cancer types ✖ No evidence

KIT p.(L576P) c.1727T>C

Relevant Therapy	FDA	NCCN	EMA	ESMO	Clinical Trials*
imatinib	✖	◑	✖	✖	✖

BAG4-FGFR1 fusion

Relevant Therapy	FDA	NCCN	EMA	ESMO	Clinical Trials*
erdafitinib	✖	✖	✖	✖	● (II)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

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Relevant Therapy Details

Current NCCN Information

● In this cancer type ○ In other cancer type ◑ In this cancer type and other cancer types

NCCN information is current as of 2022-03-31. For the most up-to-date information, search www.nccn.org.
For NCCN International Adaptations & Translations, search www.nccn.org/global/international_adaptations.aspx.

KIT p.(L576P) c.1727T>C

● imatinib

Cancer type: Cutaneous Melanoma **Variant class:** KIT L576P mutation

NCCN Recommendation category: 2A

Population segment (Line of therapy):
- Metastatic, Unresectable, Progression (Second-line therapy, Subsequent therapy); Useful in certain circumstances

Reference: NCCN Guidelines® - NCCN-Cutaneous Melanoma [Version 2.2022]

○ imatinib

Cancer type: Gastrointestinal Stromal Tumor **Variant class:** KIT exon 11 mutation

NCCN Recommendation category: 2A

Population segment (Line of therapy):
- Resectable (Neoadjuvant therapy); Preferred intervention
- Resected (Adjuvant therapy); Preferred intervention

Reference: NCCN Guidelines® - NCCN-Gastrointestinal Stromal Tumor [Version 1.2022]

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Clinical Trials in Taiwan region:

Clinical Trials Summary

BAG4-FGFR1 fusion

NCT ID	Title	Phase
NCT04083976	A Phase II Study of Erdafitinib in Subjects With Advanced Solid Tumors and FGFR Gene Alterations.	II

Alerts Informed By Public Data Sources

Current FDA Information

Contraindicated Not recommended Resistance Breakthrough Fast Track

FDA information is current as of 2022-04-13. For the most up-to-date information, search www.fda.gov.

BAG4-FGFR1 fusion

Debio 1347

Cancer type: Solid Tumor

Variant class: FGFR1 aberration

Supporting Statement:
The FDA has granted Fast Track Designation to the FGFR 1-3 inhibitor, debio 1347, for FGFR1/2/3 alterations in unresectable or metastatic solid tumors.

Reference:

https://www.debiopharm.com/drug-development/press-releases/fda-grants-fast-track-designation-to-debiopharm-internationals-debio-1347-for-the-treatment-of-patients-with-unresectable-or-metastatic-tumors-with-a-specific-fgfr-gene-alteration/

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Signatures

Testing Personnel: Shin-Ying Lin 2022.07.04 09:11:45 +08'00'

Laboratory Supervisor: 2022.07.04 09:11:57 +08'00'

Pathologist: Yi-Chu Yeh 2022.07.04 09:12:08 +08'00'

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References

1. Babina et al. Advances and challenges in targeting FGFR signalling in cancer. Nat. Rev. Cancer. 2017 May;17(5):318-332. PMID: 28303906
2. Ahmad et al. Mechanisms of FGFR-mediated carcinogenesis. Biochim. Biophys. Acta. 2012 Apr;1823(4):850-60. PMID: 22273505
3. Sarabipour et al. Mechanism of FGF receptor dimerization and activation. Nat Commun. 2016 Jan 4;7:10262. doi: 10.1038/ncomms10262. PMID: 26725515
4. Helsten et al. The FGFR Landscape in Cancer: Analysis of 4,853 Tumors by Next-Generation Sequencing. Clin. Cancer Res. 2016 Jan 1;22(1):259-67. PMID: 26373574
5. Peter et al. Comprehensive genomic characterization of squamous cell lung cancers. Nature. 2012 Sep 27;489(7417):519-25. PMID: 22960745
6. Ciriello et al. Comprehensive Molecular Portraits of Invasive Lobular Breast Cancer. Cell. 2015 Oct 8;163(2):506-19. PMID: 26451490
7. Cancer et al. Integrated genomic characterization of endometrial carcinoma. Nature. 2013 May 2;497(7447):67-73. PMID: 23636398
8. Weinstein et al. The Cancer Genome Atlas Pan-Cancer analysis project. Nat. Genet. 2013 Oct;45(10):1113-20. PMID: 24071849
9. Cerami et al. The cBio cancer genomics portal: an open platform for exploring multidimensional cancer genomics data. Cancer Discov. 2012 May;2(5):401-4. PMID: 22588877
10. Lew et al. The precise sequence of FGF receptor autophosphorylation is kinetically driven and is disrupted by oncogenic mutations. Sci Signal. 2009 Feb 17;2(58):ra6. PMID: 19224897
11. Jackson et al. 8p11 myeloproliferative syndrome: a review. Hum. Pathol. 2010 Apr;41(4):461-76. PMID: 20226962
12. Li et al. Identification of a novel partner gene, TPR, fused to FGFR1 in 8p11 myeloproliferative syndrome. Genes Chromosomes Cancer. 2012 Sep;51(9):890-7. PMID: 22619110
13. Wasag et al. The kinase inhibitor TKI258 is active against the novel CUX1-FGFR1 fusion detected in a patient with T-lymphoblastic leukemia/lymphoma and t(7;8)(q22;p11). Haematologica. 2011 Jun;96(6):922-6. PMID: 21330321
14. https://www.debiopharm.com/drug-development/press-releases/fda-grants-fast-track-designation-to-debiopharm-internationals-debio-1347-for-the-treatment-of-patients-with-unresectable-or-metastatic-tumors-with-a-specific-fgfr-gene-alteration/
15. Cha et al. FGFR2 amplification is predictive of sensitivity to regorafenib in gastric and colorectal cancers in vitro. Mol Oncol. 2018 Jun;12(7):993-1003. PMID: 29573334
16. Chae et al. Inhibition of the fibroblast growth factor receptor (FGFR) pathway: the current landscape and barriers to clinical application. Oncotarget. 2017 Feb 28;8(9):16052-16074. PMID: 28030802
17. Porta et al. FGFR a promising druggable target in cancer: Molecular biology and new drugs. Crit. Rev. Oncol. Hematol. 2017 May;113:256-267. PMID: 28427515
18. Gozgit et al. Ponatinib (AP24534), a multitargeted pan-FGFR inhibitor with activity in multiple FGFR-amplified or mutated cancer models. Mol. Cancer Ther. 2012 Mar;11(3):690-9. PMID: 22238366
19. Yamamoto et al. Lenvatinib, an angiogenesis inhibitor targeting VEGFR/FGFR, shows broad antitumor activity in human tumor xenograft models associated with microvessel density and pericyte coverage. Vasc Cell. 2014 Sep 6;6:18. doi: 10.1186/2045-824X-6-18. eCollection 2014. PMID: 25197551
20. Kim et al. Pazopanib, a novel multitargeted kinase inhibitor, shows potent in vitro antitumor activity in gastric cancer cell lines with FGFR2 amplification. Mol. Cancer Ther. 2014 Nov;13(11):2527-36. PMID: 25249557
21. Hibi et al. FGFR gene alterations in lung squamous cell carcinoma are potential targets for the multikinase inhibitor nintedanib. Cancer Sci. 2016 Nov;107(11):1667-1676. PMID: 27581340
22. Lim et al. Efficacy and safety of dovitinib in pretreated patients with advanced squamous non-small cell lung cancer with FGFR1 amplification: A single-arm, phase 2 study. Cancer. 2016 Oct;122(19):3024-31. PMID: 27315356
23. Paik et al. A Phase Ib Open-Label Multicenter Study of AZD4547 in Patients with Advanced Squamous Cell Lung Cancers. Clin. Cancer Res. 2017 Sep 15;23(18):5366-5373. PMID: 28615371
24. Ségaliny et al. Receptor tyrosine kinases: Characterisation, mechanism of action and therapeutic interests for bone cancers. J Bone Oncol. 2015 Mar;4(1):1-12. PMID: 26579483
25. Berenstein. Class III Receptor Tyrosine Kinases in Acute Leukemia - Biological Functions and Modern Laboratory Analysis. Biomark Insights. 2015;10(Suppl 3):1-14. PMID: 26309392
26. Ashman. The biology of stem cell factor and its receptor C-kit. Int. J. Biochem. Cell Biol. 1999 Oct;31(10):1037-51. PMID: 10582338

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References (continued)

27. Cardoso et al. The SCF/c-KIT system in the male: Survival strategies in fertility and cancer. Mol. Reprod. Dev. 2014 Dec;81(12):1064-79. PMID: 25359157
28. Abbaspour et al. Receptor tyrosine kinase (c-Kit) inhibitors: a potential therapeutic target in cancer cells. Drug Des Devel Ther. 2016;10:2443-59. PMID: 27536065
29. Liang et al. The C-kit receptor-mediated signal transduction and tumor-related diseases. Int. J. Biol. Sci. 2013;9(5):435-43. PMID: 23678293
30. Garcia-Montero et al. KIT mutation in mast cells and other bone marrow hematopoietic cell lineages in systemic mast cell disorders: a prospective study of the Spanish Network on Mastocytosis (REMA) in a series of 113 patients. Blood. 2006 Oct 1;108(7):2366-72. PMID: 16741248
31. Chatterjee et al. Mastocytosis: a mutated KIT receptor induced myeloproliferative disorder. Oncotarget. 2015 Jul 30;6(21):18250-64. PMID: 26158763
32. https://www.accessdata.fda.gov/drugsatfda_docs/label/2022/021588s060lbl.pdf
33. NCCN Guidelines® - NCCN-Cutaneous Melanoma [Version 2.2022]
34. NCCN Guidelines® - NCCN-Soft Tissue Sarcoma [Version 1.2022]
35. Casali et al. Gastrointestinal stromal tumours: ESMO-EURACAN Clinical Practice Guidelines for diagnosis, treatment and follow-up. Ann. Oncol. 2018 Oct 1;29(Supplement_4):iv68-iv78. PMID: 29846513
36. Jonathan. KIT Oncogenic Mutations: Biologic Insights, Therapeutic Advances, and Future Directions. Cancer Res. 2016 Nov 1;76(21):6140-6142. PMID: 27803101
37. NCCN Guidelines® - NCCN-Acute Myeloid Leukemia [Version 1.2022]
38. Lim et al. Systemic mastocytosis in 342 consecutive adults: survival studies and prognostic factors. Blood. 2009 Jun 4;113(23):5727-36. PMID: 19363219
39. Verstovsek. Advanced systemic mastocytosis: the impact of KIT mutations in diagnosis, treatment, and progression. Eur. J. Haematol. 2013 Feb;90(2):89-98. PMID: 23181448

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